

History and Opening example



Poisson

BC. - Games of chance in ancient Greece and Rome

17C - Fermat & Pascal

18C - Jacob Bernoulli derives the law of large numbers

De Moivre introduces the normal distribution and the CLT

19C - Laplace generalizes the CLT

Legendre and Gauss apply the least squares method

Poisson introduces the Poisson distribution

Chebyshev, Markov, Lyapunov Study limit theorems

20C - Kolmogorov introduces the axiomatic system for prob. theory

1D Example Counting α particles $\alpha \rightarrow \text{p p n n} : 2 \text{ protons} + 2 \text{ neutrons}$

$N_t := \#$ of clicks during the time interval $[0, t)$

One way to derive $P(N_t = r)$ is to divide $[0, t)$ into n subintervals

$$P(N_t = r) = P(r \text{ intervals with single clicks, } n-r \text{ without}) \\ + P(\text{some intervals have 2 or more clicks})$$

Assumptions

- If we choose a large n , the second probability is negligible.
- Memoryless property (each subinterval is indep. with each other)
- $P(\text{Click during } [a, b)) \propto (b-a) \frac{\mu(b-a)}{\mu(b-a)}$

$$\Rightarrow P(N_t = r) = \binom{n}{r} \left(\frac{\mu t}{n}\right)^r \left(1 - \frac{\mu t}{n}\right)^{n-r} \xrightarrow{n \rightarrow \infty} \frac{(\mu t)^r}{r!} \cdot e^{-\mu t}$$

$$\therefore N_t \sim \mathcal{P}(\mu t) \text{ with rate } E\left[\frac{N_t}{t}\right] = \mu$$

t.m.i.

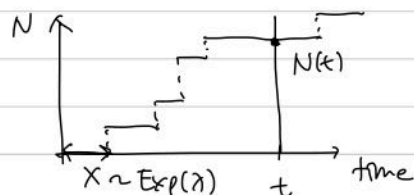
Waiting time becomes memoryless

Let $N(t) \sim \text{Poi}(\lambda t) : \#$ of events in interval $[0, t]$

and X : time until the first event

$$\Rightarrow X \sim \text{Exp}(\lambda)$$

$$(\because P(N(t) = 0) = e^{-\lambda t} = P(X \geq t) \Rightarrow \underline{F_X(x) = 1 - e^{-\lambda x}})$$



$\rightarrow$ The time from any point until the next event is $\sim \text{exp}(\lambda)$

Memoryless Property of $X \sim \text{Exp}(\lambda)$

$$P(X > s+t | X > s) = P(X > t) \rightarrow$$

$Y \sim \text{Geom}(p)$ is the only discrete r.v. with this prop.

1 Poisson Distribution $\mathcal{P}(\mu)$

- $X \sim \mathcal{B}(n, p)$: binomial distribution (n indep. trials with p)

$$\Rightarrow P(X=r) = \binom{n}{r} p^r (1-p)^{n-r}$$

$$\begin{aligned} & \downarrow n \rightarrow \infty \text{ while holding } \mu = np \text{ constant} \\ & \lim_{n \rightarrow \infty} \frac{n(n-1) \dots (n-r+1)}{r!} \cdot \left(\frac{\mu}{n}\right)^r \left(1 - \frac{\mu}{n}\right)^{n-r} \\ & = \frac{\mu^r}{r!} e^{-\mu} =: \pi_r(\mu) \end{aligned}$$

Def $X \sim \mathcal{P}(\mu) \Leftrightarrow P(X=r) = \pi_r(\mu) \text{ and } X \in \mathbb{Z}_{\geq 0}$
($r \geq 0$)

$$\begin{aligned} \bullet \mathbb{E}(X) &= \sum_{n=0}^{\infty} n \frac{\mu^n}{n!} e^{-\mu} \\ &= \mu \sum_{n=0}^{\infty} \frac{\mu^n}{n!} e^{-\mu} = \mu \end{aligned}$$

- Sometimes we extend $\mathcal{P}(\mu)$ to contain $\mathcal{P}(0)$, $\mathcal{P}(\infty)$

- Moments of X : $\mathbb{E}(X^2)$, $\mathbb{E}(X^3)$?

For $z \in \mathbb{C}$ with $|z| \leq 1$, $|z^X| \leq 1$

$$\begin{aligned} \mathbb{E}(z^X) &= \sum_{n=0}^{\infty} z^n \cdot \frac{\mu^n}{n!} e^{-\mu} = e^{\mu(z-1)} \\ \left. \begin{aligned} \frac{d}{dz} \Big|_{z=1} \mathbb{E}(X) &= \mu \\ \mathbb{E}(X(X-1)) &= \mu^2 \\ \mathbb{E}(X(X-1)(X-2)) &= \mu^3 \end{aligned} \right\} \rightarrow \begin{aligned} \mathbb{E}(X^2) &= \mu^2 + \mu \\ \mathbb{E}(X^3) &= \mu^3 + 3\mu^2 + \mu \end{aligned} \end{aligned}$$

$$\bullet \sum_{k=0}^n \pi_k(\mu) = 1 - \int_0^{\mu} \pi_n(\lambda) d\lambda \quad (\rightarrow 1 \text{ as } n \rightarrow \infty)$$

$$\begin{aligned} \therefore \frac{d}{d\mu} \pi_k(\mu) &= \frac{d}{d\mu} \left(\frac{\mu^k}{k!} e^{-\mu} \right) = \frac{\mu^{k-1}}{(k-1)!} e^{-\mu} - \frac{\mu^k}{k!} e^{-\mu} \\ &= \pi_{k-1}(\mu) - \pi_k(\mu) \end{aligned}$$

$$\left(\frac{d}{d\mu} \sum_{k=0}^n \pi_k(\mu) = -\pi_n \quad (\pi_{-1}(\mu) := 0) \right) \quad \& \text{ integrate from } 0 \text{ to } \mu$$

- (Additivity) $X \sim \mathcal{P}(\lambda)$, $Y \sim \mathcal{P}(\mu)$ & X, Y are independent
 $\Rightarrow X+Y \sim \mathcal{P}(\lambda+\mu)$

pf $P(X=r \text{ and } Y=s) \stackrel{\text{independence}}{=} P(X=r) P(Y=s) = \frac{\lambda^r}{r!} e^{-\lambda} \frac{\mu^s}{s!} e^{-\mu}$
 So, $P(X+Y=n) = \sum_{r+s=n} \frac{\lambda^r \mu^s}{r! s!} e^{-(\lambda+\mu)}$
 $= \left(\sum_{r+s=n} \binom{n}{r} \lambda^r \mu^s \right) \cdot \frac{1}{n!} \cdot e^{-(\lambda+\mu)}$
 $= \frac{(\lambda+\mu)^n}{n!} e^{-(\lambda+\mu)}$

- (Countable Additivity) $X_1, X_2, \dots$: indep & $X_i \sim \mathcal{P}(\mu_i)$

If $\sum \mu_i$ converges, $S = \sum X_i \sim \mathcal{P}(\sum \mu_i)$

If $\sum \mu_i \rightarrow \infty$, $S \sim \mathcal{P}(\infty)$

pf $S_n = \sum_{i=1}^n X_i \sim \mathcal{P}(\sum_{i=1}^n \mu_i) \quad (\text{by induction})$

$\forall r, P\{S_n \leq r\} = \sum_{k=0}^r \pi_k(\sigma_n) = 1 - \int_0^{\sigma_n} \pi_r(\lambda) d\lambda$: decreases as $n \uparrow$

Also, $\{S_1 \leq r\} \supseteq \{S_2 \leq r\} \supseteq \dots \supseteq \{S_n \leq r\}$

with $\lim_{n \rightarrow \infty} \bigcap_{k=1}^n \{S_k \leq r\} = \{S \leq r\}$

$\therefore P\{S \leq r\} = \lim_{n \rightarrow \infty} P\{S_n \leq r\} = \lim_{n \rightarrow \infty} \sum_{k=0}^r \pi_k(\sigma_n)$

If $\sigma_n \rightarrow \sigma$ as $n \rightarrow \infty$, $P\{S \leq r\} = \sum_{k=0}^r \pi_k(\sigma)$

If $\sigma_n \rightarrow \infty$ as $n \rightarrow \infty$, $P\{S \leq r\} = 0 \quad \forall r$

- (Obtaining binomial/multinomial distributions by conditioning)
 $S = X_1 + X_2 + \dots + X_n$ where $X_i \sim \mathcal{P}(\mu_i)$ & $X_1, \dots, X_n$ are independent.
 Then $S \sim \mathcal{P}(\sigma)$ where $\sigma = \mu_1 + \dots + \mu_n$

$\Rightarrow P(X_1=r_1, \dots, X_n=r_n \mid S=s)$

$= \prod_{i=1}^n \frac{\mu_i^{r_i}}{r_i!} e^{-\mu_i} \bigg/ \frac{(\sigma)^s}{s!} e^{-\sigma}$

$= \binom{s}{r_1, r_2, \dots, r_n} \cdot \left(\frac{\mu_1}{\sigma}\right)^{r_1} \cdot \left(\frac{\mu_2}{\sigma}\right)^{r_2} \dots \left(\frac{\mu_n}{\sigma}\right)^{r_n}$: multinomial distribution

For $n=2$,

$P(X=r, Y=n-r \mid X+Y=n) = \binom{n}{r} p^r (1-p)^{n-r}$ where $p = \frac{\mu_X}{\mu_X + \mu_Y}$
 $\sim \mathcal{B}(n, p)$

- Obtaining Poisson variables from binomial conditional distribution
 $X \sim \mathcal{P}(\mu)$
 $Y|X \sim \mathcal{B}(X, p) \Rightarrow Y \text{ and } X-Y \text{ are independent Poisson variables}$

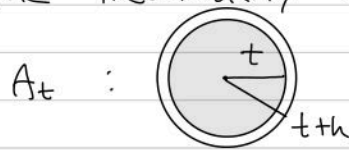
pf $P(Y=r, X=n) = P(Y=r | X=n) P(X=n)$

$$= \binom{n}{r} p^r (1-p)^{n-r} \cdot \frac{\mu^n}{n!} e^{-\mu}$$

$$= \underbrace{\frac{(\mu p)^r}{r!} e^{-\mu p}}_{\sim \mathcal{P}(\mu p)} \cdot \underbrace{\frac{(\mu(1-p))^{n-r}}{(n-r)!} e^{-\mu(1-p)}}_{\sim \mathcal{P}(\mu(1-p))}$$

(Sum over all $n \geq r$)

2 The inevitability of the Poisson distribution



$A_t : \begin{matrix} \text{shaded sector} \\ \text{unshaded area} \end{matrix} \begin{matrix} t \\ t+h \end{matrix} \rightsquigarrow \begin{matrix} p_n(t) = P\{N(A_t) = n\} \\ q_n(t) = P\{N(A_t) \leq n\} \end{matrix} \left. \begin{matrix} \text{differentiable} \\ \text{a.e.} \end{matrix} \right\} \begin{matrix} (\because \text{only jump} \\ \text{discontinuities}) \end{matrix}$

Consider $N(A_t)$ when it jumps from n to $n+1$.

$$\mu(t) = E\{N(A_t)\} = \sum_{n=0}^{\infty} n \cdot p_n(t)$$

$$P\{\exists \text{ point } \in A_{t+h} \setminus A_t \mid N(A_t) = n\} = \mu(t+h) - \mu(t)$$

$\uparrow$ Suppose this is independent of $N(A_t)$

$P\{\text{jump occurs} : q_n(t) - q_n(t+h) = p_n(t) \cdot \{\mu(t+h) - \mu(t)\}$
 given $N(A_t) = n\}$

Letting $h \rightarrow 0$, $-\frac{dq_n(t)}{dt} = p_n(t) \cdot \frac{d\mu(t)}{dt}$

From $q_0 = p_0$, $p_n = q_n - q_{n-1}$

$$-\frac{dp_0}{dt} = p_0 \frac{d\mu}{dt} \quad \text{and} \quad -\frac{dp_n}{dt} = \frac{d}{dt}(q_n - q_{n-1}) = (p_n - p_{n-1}) \frac{d\mu}{dt}$$

$\downarrow$

$$\frac{d}{dt}(\log p_0 + \mu) = 0$$

$$p_0(0) = 1, \mu(0) = 0 \Rightarrow \underline{p_0(t) = e^{-\mu(t)}}$$

$\downarrow \times e^\mu$

$$\frac{d}{dt}(p_n e^\mu) = p_{n-1} e^\mu \frac{d\mu}{dt}$$

$$p_n(0) = 0 \quad (n \geq 1)$$

$$p_n(t) = e^{-\mu(t)} \int_0^t p_{n-1}(s) e^{\mu(s)} \frac{d\mu}{ds} ds$$

Inductively, $p_1(t) = e^{-\mu(t)} \int_0^t e^{-\mu(s)} e^{\mu(s)} \frac{d\mu}{ds} ds = \mu(t) e^{-\mu(t)}$

$\vdots$

$$p_n(t) = \frac{\mu(t)^n}{n!} e^{-\mu(t)} = P(N(A_t) = n)$$

$$\therefore N(A_t) \sim \mathcal{P}(\mu(t))$$

An inevitable consequence of the 'complete randomness' inherent in our assumptions of independence. (More in Ch.8)

3 Probability Spaces for Poisson Processes

Let $(\Omega, \mathcal{F}, P)$ be a probability space.
 $\swarrow$ "Sample space" of outcomes $\searrow$ probability measure $P: \mathcal{F} \rightarrow [0,1]$
 $\downarrow$ "events" $\subset 2^\Omega$
 σ -field on Ω

Def A σ -field (or σ -algebra) on Ω is a nonempty collection $\mathcal{F}$ of subsets of Ω closed under complement, countable union, and countable intersections.

$(\Omega, \mathcal{F})$ is called a measurable space.

Def A real variable $X: \Omega \rightarrow \mathbb{R}$ is a measurable real function.
 i.e., $\{X \leq r\} \in \mathcal{F} \quad \forall r \in \mathbb{R}$

Def A random process is a collection of random variables $\{X_n\}_{n \in \mathbb{T}}$ where each $X_n \in S$ and (S, Σ) is a measurable space.
 $\swarrow$ "state space"; finite or countable

or $\Pi: \Omega \rightarrow S^\infty$ (S^∞ : countable subset of S)

If A is a 'test set' in S ,

$$N(A) = \# \{ \Pi(\omega) \cap A \} : \Omega \rightarrow \{0, 1, 2, \dots, \infty\}$$

If $N(\cdot)$ is measurable for any A , $N(A)$ are random variables.
 i.e., $\{\omega: N(A) = n\} \in \mathcal{F}$

ex $S = \mathbb{R}$, $A = (a, b) \Rightarrow$ Any open set $G = \overset{\text{disjoint}}{\bigcup_{j=1}^{\infty}} A_j$, $N(G) = \sum_{j=1}^{\infty} N(A_j)$ is a r.v.
 Any closed set $F = \bigcap_{i=1}^{\infty} G_i$ ($G_i \supset G_{i+1}$),
 $N(F) = \lim_{i \rightarrow \infty} N(G_i)$
 $\rightarrow$ For any Borel set A , $N(A)$ is a random variable

ex $S = \mathbb{R}^2$, $A = (a, b) \times (c, d)$ (... measure theory ...)

< To be continued >

Section 2.1.

- Formal definition of a Poisson process $\Pi \in S^\infty$
 where $N(A)$ are indep. ($N(A_1), N(A_2)$ are indep if $A_1 \cap A_2 = \emptyset$)
 $N(A) \sim \mathcal{P}(\mu(A))$ where μ is a non-atomic measure on S

Section 2.5

- Conversely, for any non-atomic measure μ on S , is there Π s.t.
 $N(A) \sim \mathcal{P}(\mu(A))$?

Explicit construction is given with easily satisfied conditions (S, μ)

$\rightarrow$ A theory of great generality and wide applicability